Sheng Zhao

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I am currently involved in VR-related research: computer vision(CV) & human computer interaction(HCI), including social computing, authentication, data visualization in VR as well as 3D generation.

Education

Tsinghua University August 2021 – Present

Mathematics and Physics&Electrical Engineering and Automation Double Major

I am currently a junior with a GPA of 3.85/4 and a major GPA of 3.89/4. You can find my transcript here.

- Physics of Optics&Quantum Mechanics(100), Physics of Electromagnetism(98), Calculus A1&A2(98), Mathematical Physics Equations(98), Computer Program(98), Linear Algebra(92).

For more information, please visit my personal website.

Research Experiences

Avatar Facial Expressions April 2023 – October 2023

Research Assistant at Tsinghua University, Beijing, China (Advisors: Yuanchun Shi&Xin Yi)

This is a research project in the field of human-computer interaction (HCI). We aim to investigate the impact of the intensity of facial expressions in VR face-to-face communication on user experience. My works include:

- Implementing VR face-to-face communication using TCP-based socket programming and the Oculus SDK, designing interactive UI to allow users to adjust facial expressions (blendshapes) themselves.
- Designing both qualitative and quantitative research strategies based on semi-structured interviews and questionnaires, following Interaction Design
- Leading experimenter in user research experiments.Initial experimental results were not satisfactory, as users felt a uncanny valley effect when facial blendshapes were magnified. I proposed several solutions based on relevant research to improve the experimental design.

This project is currently in the stage of data analysis and paper writing, and I plan to submit it to CSCW as co-first authors. You can find the platform source code and video demo here.

VR Implicit Authentication September 2023 – Present

Research Assistant at Tsinghua University, Beijing, China (Advisor: Xin Yi)

This is a research project in the field of human-computer interaction and application security. We aim to explore the feasibility and security of identity authentication based on different sequences of hand, head, and facial movements. My works include:

- Reviewing multiple papers on head movements, eye movements, and biometric authentication to propose the current authentication framework. Here are papers that inspired me to propose the authentication algorithm.
- Using Unity and Quest Pro to extract a series of data related to hand, head, and facial movements as a test and training set for neural networks.
- A framework of authentication by machine learning, using SVM, KNN, RFS etc. to recognize the content and pattern of each user.

You can find the source code here.

3D Generation by diffusion model October 2023 – Present

Visiting Student of C3I Lab at Tsinghua University, Beijing, China (Advisor: Bowen Zhou)

This is a research project in the field of computer vision. My works include:

- Reviewing multiple papers and learning knowledges on diffusion model.
- Introducing the VRDiffuser, a conditional generative model for 3D VR facial generation.

Taxonomy of embodied interactions October 2023 – Present

Remote Research Assistant of GTVis Lab at Georgia Institute of Technology, USA

This is a research project in the field of human-computer interaction.Just investigating embodied interactions in immersive analytics. My works include:

- Reviewing academic papers and joining Brainstorming
- Developing Demos of embodied interactions for data visualization

Science and Technology Innovation Competition

Tsinghua University Software Design Competition December 2022 – January 2023

Developed a Smarthome Android app using Java, which includes features like Bluetooth connectivity, weather forecast queries, and communication with hardware devices. Finally I won the Second Prize. The project has been open-sourced here.

The 12th Tsinghua University Digital System Design Competition August 2022 – October 2023

Designed a small car driven by MSP430 with various sensors for autonomous navigation and resource acquisition. Finally I won the First Prize(1/60). The project source code can be found here.

Skills

- Wizard in C#(NET,Socket,Unity3D,VR)
- Wizard in Python(NumPy,Pandas,Matplotlib,Scikit-learn,TensorFlow)
- Proficient in Java(Android,Web spider)
- Proficient in machine learning, deep learning, computer vision(SVM, KNN, RF, CNN, MLP, ResNet, Diffusion, Transformer).
- Proficient in user experience design(Qualitative research, Quantitative research)
- Skilled in FPGA, STM32, MSP430(Hardware design)
- Proficient in using Git, LaTeX, and Markdown.

Awards

Provincial First Prize, 36th China Physics Olympiad (CPHO) September 2020
First Prize, The 12th Tsinghua University Digital System Design Competition October 2022
Second Prize, Tsinghua University Sports Excellence Scholarship October 2022
Second Prize, Tsinghua University ”Smart Coastal” Software Design Competition January 2023
First Prize, Tsinghua University Scientific And Technological Innovation Scholarship October 2023
First Prize, Tsinghua University Sports Excellence Scholarship October 2023
First Prize, The 3rd Tsinghua Craftsman Competition (Presented a representative speech) November 2023